

PHONOLOGY

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1 Basic terminology

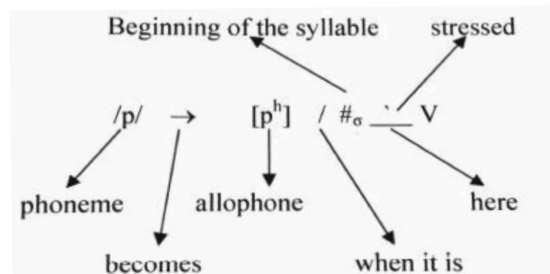
- if two sounds in a language contrastive, these sounds are different **phonemes**
 - a phoneme is an abstract unit that corresponds to a set of similar speech sounds called **allophones**
 - a phoneme can distinguish one word from another in a language
- in order to determine that two sounds are separate phonemes:
 - a **minimal pair** is two words that differ in one sound or phonological element, e.g. *pat* vs. *bat*
 - a **near-minimal pair** is two words where the immediate environment of the contrastive sounds are exactly the same, e.g. *this* [ðɪs] vs. *thistle* [θɪstl], /ð/ and /θ/ show up in the context #_ɪ
 - these means that the two sounds are **contrastive**
- in order to determine that two sounds are allophones of the same phoneme:
 - the list of environments that the sounds occur in are different, e.g. [p] and [p^h] in English
 - this means that the two sounds are in **complementary distribution**
 - the **elsewhere case** is the phonetic representation of the underlying phoneme
- slashes / / represent phonemes while square brackets [] represent allophones
- if a feature distinguishes two phonemes, it is a **distinctive feature**
 - this can vary across languages, e.g. aspiration is distinctive in Hindi but not English
 - a set of phonemes that share certain distinctive features is called a **natural class**
- **phonological rules** capture the distribution of allophones based on features of the environment

2 Natural classes

- a natural class is a set of all sounds in a language that share certain distinctive features
- since distinctive features differ across languages, natural classes also differ across languages
- features of vowels and consonants are all potential natural classes (see phonetics handout):
 - the different vowel heights, backness, roundedness, tenseness
 - the different consonant places of articulation, manners of articulation, voicing
- additional natural classes include:
 - **vocalic** (vowels and glides) vs. **consonantal** (consonants)
 - **sonorant** (voiced vowels, nasals, trills, taps/flaps, liquids, glides) vs. **obstruent** (stops, fricatives, affricates, voiceless sounds)
 - **coronal** (interdental, dental, alveolar, post-alveolar, retroflex consonants)
 - **continuant** (vowels, fricatives, trills, tap/flaps, liquids, glides)
- the sonorant vs. obstruent distinction refers to how much airflow is restricted in the sound

3 Phonological rules

- phonological rules capture the distribution of allophones based on features of the environment



- there are common classes of phonological rules
 - **assimilation** occurs when a sound changes to become more similar to other nearby sounds
 - * e.g. nasal place assimilation: /n/ → [αplace] / _[αplace]
 - * e.g. voice assimilation: /p/ → [b] / V_V
 - **dissimilation** occurs when a sound changes to become less similar to other nearby sounds
 - * e.g. /t/ → [s] / _[-son, -cont]
 - **deletion** occurs when a sound is deleted
 - * e.g. /g/ → ∅ / _[+nas]
 - **insertion (epenthesis)** occurs when a sound is added
 - * e.g. ∅ → [i] / C_C
 - **metathesis** occurs when two sounds exchange places
 - * e.g. V₁V₂C → V₁CV₂ / _#
- phonological rules often need to follow a specific order with respect to each other
- the change that occurs in a phonological rule is often a feature that is related to its environment
- **phonotactics** govern which sound sequences are permissible, and where

4 Syllable structure

- a **syllable** is a unit of organization for a sequence of speech sounds, the “building blocks” of words
 - a syllable can be broken down into an **onset** (initial consonants) and a **rhyme**
 - a rhyme can be further broken down into a **nucleus** (vowels) and a **coda** (final consonants)
- to decide whether a consonant is an onset or coda, **maximize the onset** when possible

